



OOps Problem Bank

Legend

★ = Core Placement Problem (Mandatory)

These problems should be completed by every student before moving to Map, Filter, and Reduce.

Section 1: Classes and Objects Basics

★ Create a Simple Class and Object

★ Class with `__init__` Constructor

★ Create Multiple Objects of a Class

Class with Default Parameter Values

★ Instance Variables vs Class Variables

Object Counter using Class Variable

Class with Multiple Methods

Section 2: Methods and the self Keyword

★ Instance Method to Display Object Details

★ Method to Update Instance Variables

Method that Returns a Calculated Value

★ Class with Methods Calling Other Methods

Method Overloading using Default Arguments

Class Method using `@classmethod`

Static Method using `@staticmethod`

Section 3: Constructors and Destructors

★ Parameterized Constructor

Default (No-Argument) Constructor

★ Constructor Overloading using Default Values

Destructor using `__del__`

Constructor Chaining with `super()`

Section 4: Encapsulation

★ Private Variables using Double Underscore

★ Getter and Setter Methods

Protected Members using Single Underscore

★ Bank Account Class with Encapsulated Balance



Property Decorator (@property) Usage

Validate Data using Setter Methods

Section 5: Inheritance

★ **Single Inheritance - Parent and Child Class**

★ **Method Overriding in Child Class**

★ **Using super() to Call Parent Constructor**

Multiple Inheritance - Two Parent Classes

★ **Multilevel Inheritance (Grandparent-Parent-Child)**

Hierarchical Inheritance (One Parent, Multiple Children)

Calling Parent Class Method from Child Class

Section 6: Polymorphism

★ **Method Overriding for Polymorphic Behavior**

★ **Polymorphism with Function and Multiple Classes**

Operator Overloading using __add__

★ **Operator Overloading using __str__ and __repr__**

Duck Typing Example

Polymorphism with Class Methods of Same Name

Section 7: Abstraction

★ **Abstract Class using ABC Module**

★ **Abstract Method Implementation in Child Classes**

Interface-Like Behavior using Abstract Base Class

Abstract Class Preventing Direct Object Creation

Section 8: Special (Dunder) Methods

★ **__str__ Method for Readable Object Printing**

__repr__ Method for Debug Representation

★ **__len__ Method for Custom Length Behavior**

__eq__ Method for Object Comparison

__add__ Method for Custom Object Addition

__getitem__ for Indexing Custom Objects

Section 9: Class Relationships

★ **Aggregation - Class Containing Another Class Object**

★ **Composition - Object Created and Owned by Another Class**

Class Containing a List of Objects



Many-to-One Relationship Modeling (Students and School)

Section 10: Real World OOP Applications

★ **Student Management System using Classes**

★ **Employee Management System using Classes**

Library Management System (Books and Members)

★ **Bank Account System with Deposit/Withdraw Methods**

Shape Area Calculator using Inheritance (Circle, Square, Rectangle)

★ **Vehicle Rental System using OOP**

E-Commerce Product and Cart System using Classes

★ **Inventory Management System using OOP**